

Dark chocolate improves coronary vasomotion and reduces platelet reactivity.



BACKGROUND: Dark chocolate has potent antioxidant properties. Coronary atherosclerosis is promoted by impaired endothelial function and increased platelet activation. Traditional risk factors, high oxidative stress, and reduced antioxidant defenses play a crucial role in the pathogenesis of atherosclerosis, particularly in transplanted hearts. Thus, flavonoid-rich dark chocolate holds the potential to have a beneficial impact on graft atherosclerosis. **METHODS AND RESULTS:** We assessed the effect of flavonoid-rich dark chocolate compared with cocoa-free control chocolate on coronary vascular and platelet function in 22 heart transplant recipients in a double-blind, randomized study. Coronary vasomotion was assessed with quantitative coronary angiography and cold pressor testing before and 2 hours after ingestion of 40 g of dark (70% cocoa) chocolate or control chocolate, respectively. Two hours after ingestion of flavonoid-rich dark chocolate, coronary artery diameter was increased significantly (from 2.36 $\pm$ 0.51 to 2.51 $\pm$ 0.59 mm, $P<0.01$), whereas it remained unchanged after control chocolate. Endothelium-dependent coronary vasomotion improved significantly after dark chocolate (4.5 $\pm$ 11.4% versus -4.3 $\pm$ 11.7% in the placebo group, $P=0.01$). Platelet adhesion decreased from 4.9 $\pm$ 1.1% to 3.8 $\pm$ 0.8% ($P=0.04$) in the dark chocolate group but remained unchanged in the control group. **CONCLUSIONS:** Dark chocolate induces coronary vasodilation, improves coronary vascular function, and decreases platelet adhesion 2 hours after consumption. These immediate beneficial effects were paralleled by a significant reduction of serum oxidative stress and were positively correlated with changes in serum epicatechin concentration.